

Xiangyu Chen

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Education

Nanyang Technological University, Singapore 2025 – Present

Ph.D. in Mechanical and Aerospace Engineering (MARS Lab). Advisor: Prof. Jianfei Yang.

Shandong University, Jinan, China

Sep 2020 – Jun 2023

M.Eng. in Control Science and Engineering. GPA: 89.4/100 (Rank 1/116). National Scholarship (top 1%).

University of Jinan, Jinan, China

Sep 2016 – Jun 2020

B.Eng. in Electrical Engineering and Automation. GPA: 89.7/100 (Rank 4/403). National Scholarship (top 1%), Government Scholarship (top 1%), Song Jian Scholarship (top 0.1%).

Standardized Tests: TOEFL 106 (R29 / L27 / S23 / W27); GRE 319 + 3.5.

Research Interests

Robotic manipulation, dexterous manipulation, embodied AI, and imitation/reinforcement learning for robot policy learning.

Publications

Bold denotes the author. ([†] co-first author where applicable.)

- [1] **X. Chen**, Y. Hu, C. Zhou, J. Yang. “Rethinking Implicit Spatial Representation in Visuomotor Policy Learning.” *arXiv:2606.15232*, 2026.
- [2] J. Li, Y. Hu, H. Geng, **X. Chen**, C. Zhou, Z. Cui, J. Yang. “M³A Policy: Mutable Material Manipulation Augmentation Policy through Photometric Re-rendering.” *IEEE/CVF CVPR (Findings)*, 2026.
- [3] J. Jia, T. An, Y. Hu, G. Li, J. Li, B. Hou, **X. Chen**, J. Bai, B. Lyu, J. Yang. “MARS Policy: Multimodality Only When It Matters.” *arXiv:2605.29766*, 2026.
- [4] J. Bai, J. Jia, Y. Hu, G. Li, **X. Chen**, T. An, K. Zuo, J. Yang. “FLASH: Efficient Visuomotor Policy via Sparse Sampling.” *arXiv:2605.15492*, 2026.
- [5] J. Jia, G. Li, **X. Chen**, T. An, Y. Hu, J. Li, X. Guo, J. Yang. “A2A: Action-to-Action Flow Matching for Robot Policy Learning.” *Robotics: Science and Systems (RSS)*, 2026.
- [6] **X. Chen**, C. Zhou, Y. Liu, J. Yang. “RM-RL: Role-Model Reinforcement Learning for Precise Robot Manipulation.” *IEEE Int. Conf. on Robotics and Automation (ICRA)*, 2026.
- [7] **X. Chen**, F. Yang, C. Wang. “iA*: Imperative Learning-based A* Search for Pathfinding.” *IEEE Robotics and Automation Letters (RA-L)*, 2025.
- [8] C. Wang, **X. Chen**, C. Li, R. Song, Y. Li, M. Q.-H. Meng. “Chase and Track: Toward Safe and Smooth Trajectory Planning for Robotic Navigation in Dynamic Environments.” *IEEE Trans. on Industrial Electronics*, 70(1):604–613, 2022.
- [9] **X. Chen**, J. Liu, J. Wu, C. Wang, R. Song. “LoPF: An Online LiDAR-Only Person-Following Framework.” *IEEE Trans. on Instrumentation and Measurement*, 71:1–13, 2022.
- [10] **X. Chen**, Y. Wang, C. Wang, R. Song, Y. Li. “Low-drift LiDAR-only Odometry and Mapping for UGVs in Environments with Non-level Roads.” *IEEE/RSJ IROS*, pp. 13174–13180, 2022.
- [11] Y. Wang, M. Xu, **X. Chen**, X. Zhang, C. Wang, R. Song. “Information Map Prediction based on Learning Network for Reinforced Autonomous Exploration.” *IEEE ROBIO*, pp. 1982–1988, 2022.

Experience

Graduate Research — MARS Lab, Nanyang Technological University

2025 – Present

- Embodied AI agents for life science: robotic and dexterous manipulation, and visuomotor policy learning that pairs foundation models with robotic arms for Chem/Bio lab automation.

Research Intern — MegaRobo, Beijing, China

Nov 2024 – Jan 2025

Research Intern — Realman, Beijing, China

May 2024 – Aug 2024

Research Intern (Remote) — SAIR Lab

Sep 2023 – Sep 2024

- Imperative-learning-based A* pathfinding; contributed Spline and ICP modules to the open-source PyPose library.

Graduate Research — Shandong University

2020 – 2023

- LiDAR SLAM and localization on non-level roads, safe and smooth trajectory planning, and LiDAR-only person-following for mobile and quadruped robots.

Research Intern — Yobotics, Jinan, China

May 2020 – Jul 2020

Honors & Awards

- National Scholarship for Graduate Students (top 1%), 2023
- CUBA (Shandong) — Third Place, 2022
- National Scholarship for Undergraduates (top 1%), 2020
- Song Jian Scholarship (top 0.1%), 2020
- Meritorious Winner, Mathematical Contest in Modeling (MCM/ICM), 2019

Professional Service

Reviewer: IEEE Transactions on Robotics (T-RO); IEEE Robotics and Automation Letters (RA-L); International Journal of Robotics Research (IJRR); Conference on Robot Learning (CoRL).